A full volume theorem under complete-radius balance

Original source and complete geometry

Fixed certified simple-white plat; prescribed knot-producing parity.
Exact shared-face coefficient, marked zeros, both level parities.

$$\lim_{N\to\infty} \frac{\log Z_{D,N}}{N} \quad = \quad \max_{q\in\mathcal{Q}_D} \sum_e h(q_e) \quad = \quad \frac{\mathrm{Vol}(Y_D)}{2\pi}$$

Entire CROSS filling period: proved

Nonzero geometric saddle; all original Fourier charges and whole-domain endpoint currents controlled.

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |I_{\mathrm{cross},N}| \quad = \quad \mathrm{Vol}(S^3 \setminus K_D(\mathbf{p}))$$

Finite sufficient condition on the same complete packing

Each port has face curvatures (x, b, c, y), with $A = x + y > 0$.

$$(A + b)(A + c) \quad \geq \quad \max\{b^2, c^2\}$$

A simpler sufficient test is $|b - c| \leq x + y$.
Protected zero curvatures remain exactly zero.

Full original invariant: proved under that condition

The LEFT/RIGHT remainder has a strict global exponential gap.

$$\lim_{N\to\infty} \frac{2\pi}{N} \log |K_N(K_D(\mathbf{p}))| \quad = \quad \mathrm{Vol}(S^3 \setminus K_D(\mathbf{p}))$$

Every independently long fixed signed twist vector;
threshold depends on D. Twists are fixed before $N \to \infty$.

Open: prove broad coverage at arbitrary bridge width

Construct actual marked packings satisfying balance for a family at each fixed width $k \geq 2$, or find a wider estimate.

A wide presentation or numerical sample does not prove coverage.